Stage-Aware Diagnosis of Diabetic Retinopathy via Ordinal Regression

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Abstract: Diabetic Retinopathy (DR) is a prominent contributor to avoidable vision loss. It leads to irreversible adverse effects when diagnosis and treatments at later stages. On-time detection, with methodical screening, prevents any visual impairment. In this work, we present a state-of-the-art ordinal regression detection framework for DR detection. The model is evaluated on the APTOS-2019 retinal fundus dataset. For enhancing critical retinal features, we use a standard and effective preprocessing pipeline, comprising of Green Channel Extraction, Noise suppression, and CLAHE. Model evaluation is centered on the Quadratic Weighted Kappa(QWK) metric. QWK captures alignment with expert clinical grading, pivoted on agreement between the actual clinical grading labels and the model inference. The proposed ordinal regression model achieves a 0.8992 QWK, surpassing existing results on the APTOS benchmark.
Keywords: Diabetic Retinopathy · Ordinal Regression · Quadratic Weighted Kappa · APTOS.

1 INTRODUCTION

In recent times, "Diabetic Retinopathy" (DR) has become a top challenge in the field of ophthalmologic diagnosis. DR cases make up a significant proportion amongst adult population around the world, in the global cases of preventable blindness [1]. DR is a retinal vascular complication of Type-II Diabetes Mellitus(T2DM), which causes progressive damage and obscureness to the eye's microcirculation.

Vision starts to deteriorate, which is staged through pathological mechanisms such as capillary leakage, retinal oedema, and aberrant neovascularization. The disorder develops because of sustained hyperglycemia, where chronically elevated blood glucose levels damage the retinal vasculature. The progressive damage advances to a spectrum of retinal complications, from mild vascular irregularities to profound vision loss.

DR affects about 103.12 million adults globally. This number is expected to rise to 160.50 million (by 2045) [2]. The prevalence of Diabetes globally is 22.27%, which varies regionally from 13.37% in Southern and Central America, to 35.9%

in Africa [2]. The numbers reflect the diabetes pandemic on a global scale, which calls for effective screening and management, to prevent and allay irreparable effects.

DR is detected via a series of stages, each visible through multiple alterations and increasing vision damage. In the initial stage, mild non-proliferative diabetic retinopathy (NPDR) is marked by small areas of vascular swelling and the presence of microaneurysms within the retina. As the condition advances to moderate NPDR, these vascular abnormalities become obvious, and in the severe NPDR stage, extensive occlusion of retinal vessels happens. Ultimately, the disease progresses to proliferative diabetic retinopathy (PDR), characterized by the growth of pathological neovascularization into the vitreous cavity, resulting in vascular leakage, progressive visual decline, and potentially complete blindness.

Fig.1. Stages of Diabetic Retinopathy (per ICDRSS)

1.1 The APTOS Dataset

The “Asia Pacific Tele-Ophthalmology Society” (APTOS) 2019 Blindness Detection dataset [3] is a dataset for the detection of diabetic retinopathy. The dataset is made up of 3,622 high-quality color fundus photographs, collected in rural areas under diverse imaging conditions. The images provide a summarized view of challenges in real-world screening. All images in the dataset are independently examined and annotated by qualified ophthalmologists classified under the “International Clinical Diabetic Retinopathy Disease Severity Scale” (ICDRSS) [4]. The scale ensures high-quality ground truth annotations for model training and evaluation.

The dataset is classified under the ICDRSS framework, which classifies DR into five ordered severity levels: from no DR (Class 0) to proliferative DR (Class 4) (Fig. 1). This hierarchical classification sheds light on the advancing stages of diabetic retinopathy. The dataset enables the construction and validation of ML models, for advancing automated disease detection. It is significant for improving diagnostic research with limited access to specialized ophthalmic care.

1.2 Related Works

Recent research sheds light on the effectiveness of transfer learning, using pre-trained convolutional neural networks(CNN) for DR diagnostic classification. Dixit et al.[5] proposed a novel approach utilizing EfficientNet-B3 with squeeze-and-excitation (SE) blocks, achieving 88.44% accuracy on the APTOS-2019 dataset. The integration of SE blocks helped to power the model, focusing on crucial features by implementing channel-wise attention mechanisms.

The fusion of attention mechanisms is a progressing trend in medical diagnosis, including DR classification. Farag et al.[6] combined DenseNet-169 with Convolutional Block Attention Module (CBAM), which enhances discriminative power. 97% accuracy in binary classification and 82% accuracy in severity grading on the ICDRSS scale were scored by the model. In summation, the CBAM refocused the model's capability on key features, filtering out distractions.

Lalithadevi et al. [7] created OptiDex, by integrating NASNet-Mobile with enhanced Cat Swarm optimization and xAI. This approach achieved high accuracy on the test set and helped in model inference reasoning. It was a major milestone in transparent and explained decision-making in medical applications, in contrast to the DL based black-box predictions.

Bodapati et al.'s [8] self-adaptive ensemble for DR achieved an accuracy of 86.22% and 0.8965 QWK. Their dual-attention model for lesion attention, combined with spatial correlation learning, proved an effective self-adaptive meta learner.

Oulhadj et al. [9] used deformable registration with a multi-CNN voting ensemble for DR stages grading. The approach worked on aligning the fundus image and background space influence, improving classification accuracy by image standardization. The work got a 0.75 QWK score.

1.3 Research Gaps

DR Detection has made significant progress within the multi-label image classification domain. Yet many challenges are yet to be solved,

1. Complexity of Feature Extraction: DR involves subtle retinal changes like hemorrhages and microaneurysms. Irrelevant features need to be discarded, with focus on the pertinent ones.
2. Imbalanced Class Count: DR Datasets are imbalanced. Normal cases dominate over severe stages and are underrepresented. It degrades the classification quality, primarily for the severe stages.
3. Preprocessing Standardization: Model scoring metrics are highly influenced by the data preprocessing step. Standardized pipelines are needed for consistent results across datasets and imaging conditions.
4. Standardization of the Kappa Metric within Healthcare: Accuracy alone is insufficient in medical AI. The Kappa metric penalizes prediction errors; it is a reliable choice for critical deployments.

The proposed work uses a ResNet-based Ordinal Regression Model to capture the pertinent Fundus features. The study combines a standardized preprocessing pipeline with a robust Deep Learning model. Green Channel Isolation, CLAHE enhancement, median filtering for noise reduction, and standardized resizing with 3-channel recreation for optimal input quality across datasets are used. The work focuses on achieving leading-edge performance metrics, using quadratic weighted kappa as the primary evaluation metric. The particular emphasis is put on improving performance in minority classes (severe DR stages).

2 METHODOLOGY

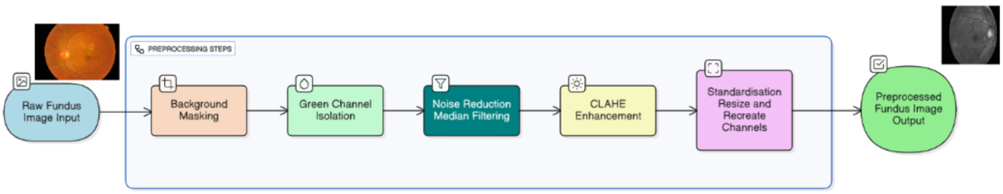

Fig.2. Preprocessing Pipeline

For standardizing input images and enhancing relevant features for DR pathology (including lesions, macular edema, hemorrhages, aneurysms, etc), a uniform preprocessing pipeline is applied to every image before it is fed into the models (Figure 2) (Algorithm 1).

The pipeline comprises:

- Background Masking: Removes the black background to isolate the region of interest (ROI).
- Green Channel Isolation: Extracts the green channel, which best highlights retinal blood vessels and pathological features, improving DR signal clarity. The green channel provides the best contrast for viewing retinal vasculature and lesions, such as microaneurysms and hemorrhages.
- Median Filtering: A median filter is applied to the isolated green channel, isolating salt-and-pepper noise and other minor artifacts, without blurring the edges of important pathological features.
- CLAHE (Contrast Limited Adaptive Histogram Equalization): It divides the whole image into separate parts and enhances contrast in each part individually. It is effective at making micro aneurysms, hard exudates, and other more prominent, subtle lesions, helping in effective, clearer grading.
- Standardization by Resizing: The single-channel enhanced image is replicated across three channels to create a 3-channel grayscale image. The input shape requirements of pre-trained models (ResNet, ViT) are 3-channel; hence, it helps to accommodate that requirement. Finally, all images are resized to a standard dimension of 224x224 pixels.

Algorithm 1: DR- Image Preprocessing Pipeline

Input: Raw Retinal Fundus Image I_{raw}

Output: Enhanced 3-Channel Image I_{proc}

$I_{res} \leftarrow$ Resize I_{raw} to 224 × 224 and convert to RGB;

$M \leftarrow$ Generate circular mask via grayscale thresholding;

$I_{masked} \leftarrow$ I_res $\odot$ M; // Remove non-retinal background

$C_G \leftarrow$ Extract Green Channel from I_{masked};

$C_{denoised} \leftarrow$ MedianFilter(C_G, kernel = 5);

$C_{enhanced} \leftarrow$ CLAHE($C_{denoised}$); // Enhance lesion visibility

$I_{proc} \leftarrow$ Stack($C_{enhanced}$, $C_{enhanced}$, $C_{enhanced}$);

return I_{proc};

2.1 Model Architecture

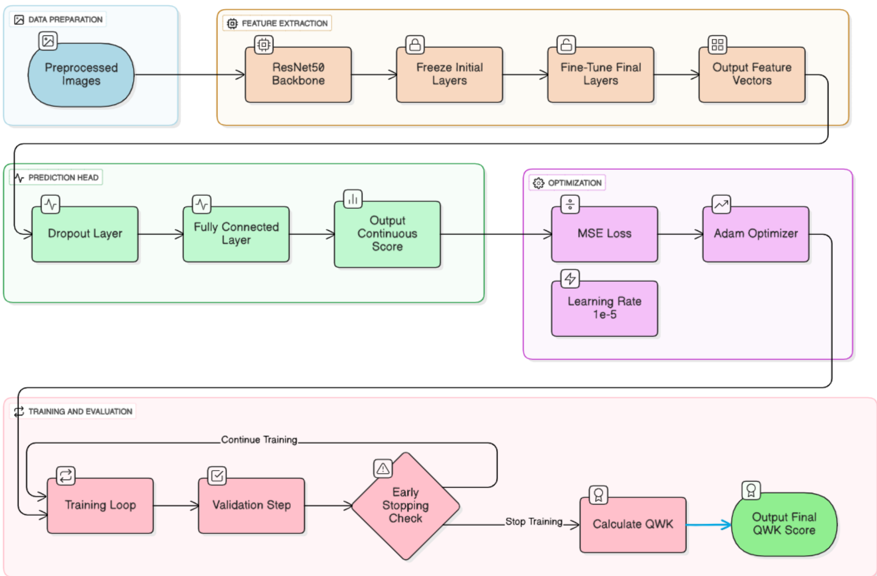

Fig.3. Ordinal Regression with ResNet50 Model Architecture

The methodology treats the problem as an ordinal regression task, using a standard ResNet50 architecture [10] as the backbone (Figure 3). A pre-trained ResNet50 model is adapted by replacing its final classification layer with a new head comprising a Dropout layer and a single Dense output neuron. The model is trained to predict a continuous severity score. A key aspect of the training strategy involves fine-tuning only the final layers of the ResNet50 model while keeping the earlier layers frozen. The Ordinal

Regression model used a higher, yet cautious, fine-tune LR 10^{-5}. The model's was trained for 30 iterations of batches of 16, using the Adam Optimizer and MSE loss. Only the final two layers (*model.layer4* and *avgpool*) of the ResNet50 backbone were unfrozen for gradient updates, minimizing the risk of catastrophic forgetting across the deeper, foundational layers. Mean Squared Error (MSE) is used to train the model on continuous outputs. For classification, the model's continuous output is clamped to a fixed range with nearest integer rounding. The output is used to determine the predicted DR stage. Model robustness relies largely on fine-tuning, with tightly constrained learning rates (LR ≤ 10^{-5}). The strategy allows the network to gradually align ImageNet-derived feature representations[11] with retinal pathology. At the same time, it preserves the integrity of the pre-trained parameter space (Algorithm 2).

Algorithm 2: Ordinal Regression Training and Inference

Input: Preprocessed training set D_{train}, weights W_{ResNet}
Output: Discrete severity grade $\hat{y} \in \{0, 1, 2, 3, 4\}$

```
// Model Initialization
Load ResNet-50 with W_ResNet weights
Replace the final fully connected layer with a Linear Layer (f: ℝ^d → ℝ^1)
Freeze all layers except layer4 and the new Linear head

// Training Phase
for each epoch e ∈ {1, …, E} do
    for each batch (X, y) do
        V_pred ← Model(X)              // Continuous scalar output
        L = (1/N) Σ(V_pred − y)^2      // Mean Squared Error Loss
        Update W using Adam optimizer (η = 10^−5)

// Inference Logic (Thresholding)
V_raw ← Model(X_test)
V_clipped ← max(0, min(4, V_raw))
ŷ ← ⌊V_clipped + 0.5⌋                 // Round to nearest integer
return ŷ
```

3. RESULTS AND DISCUSSION

Accurate Classification of DR requires an evaluation metric that reflects the clinical severity of misdiagnosis. Due to the ordinal character of the DR grading scale (stages 0 through 4), standard metrics like classification accuracy or the F1-score are insufficient, as they see all misclassifications equally. The central metric for this comparative study is the Quadratic Weighted Kappa (QWK). The QWK metric is a score-weighted variation of Cohen's Kappa, which measures inter-rater reliability (Eq. 1). It also integrates the disagreement weights between model's predicted scores and truth labels assigned under the dataset[12].

$$QWK = 1 - \frac{\sum_{i,j} W_{i,j} O_{i,j}}{\sum_{i,j} W_{i,j} E_{i,j}} \quad (1)$$

The ResNet50-based Ordinal Regression model achieved the highest ever performance, registering a peak Validation Quadratic Weighted Kappa (QWK) of 0.8992, with a Validation Mean Squared Error (MSE) loss of 0.3871 (Figure 4).

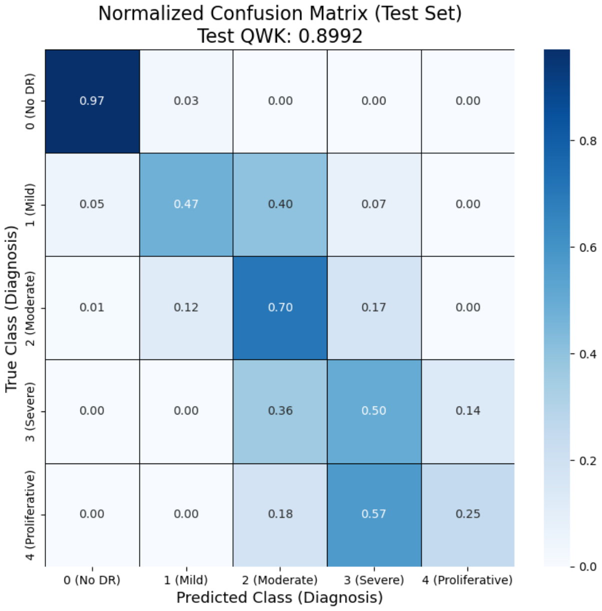

Fig.4. Ordinal Regression Test Set Confusion Matrix

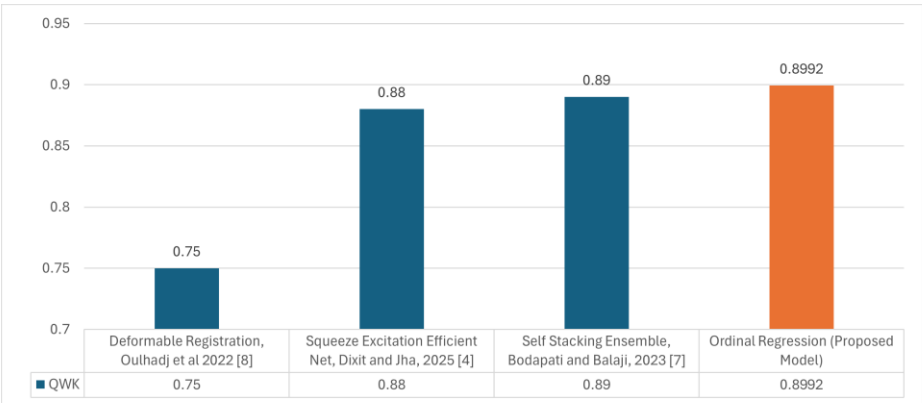

Fig.5. Comparative OWK between Related Works and Proposed Model

Figure 5 clearly illustrates the model's high QWK (orange bar in the Figure), compared with recent works in DR detection and classification. The novel Ordinal Regression with a ResNet50-based model performed better. The incorporation of a Fundus Aligned pipeline resulted in a better-performing model on the APTOS 2019 dataset for DR 5-stage diagnostic classification.

The Regression-based architecture used Mean Squared Error (MSE) loss. The model gets the top performance tier with a QWK score of 0.8992. Its advantage over classification-based models stresses the alignment between regression-oriented optimisation and the QWK metric. The approach penalises larger ordinal deviations more effectively. It thereby validates our model's industrial and clinical-grade reliability and the usefulness of the unified preprocessing pipeline.

4. CONCLUSION AND FUTURE WORK

This study explored applications of distinct machine learning and deep learning paradigms to the clinical task of DR grading of fundus images, where both accuracy and error severities contribute to diagnostic implications. The proposed approach formulates the problem as an Ordinal Learning task over a flat multi-label classification program. The methodology proves to have a structured progression of disease severity, crucial for clinical applications.

The work developed a distinct machine learning paradigm for the ordinal classification of DR into five severity grades (0 to 4). A unified preprocessing pipeline comprising of Green Channel Isolation, CLAHE, and Median Filtering was applied on the data. The performance of the Ordinal Regression+ResNet50 model was evaluated using the Quadratic Weighted Kappa (QWK) metric. "QWK" accounts for the severity of misclassification errors, inherent in ordinal grading. The model surpassed the ICDRSS QWK threshold of 0.8.

Although the results were encouraging, several limitations were identified. The processor's bulkiness added substantial overhead, which extended training time. It also fell short in fully capturing intricate, stage-specific features. More work is required to identify optimal preprocessing methods for Fundoscopy feature extraction.

The diagnostic transparency of the results can be elaborated by Explainable AI, helping diagnosticians and medical professionals in decision-making. Improving the interpretability of medical models builds trust in automated grading, and critical clinical deployments.

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